$^{2}H(^{34}S,n\gamma)$ 1973Wa10

$^{34}S(d,n)^{35}Cl$ in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200-μg/cm^2 TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35-cm^3 Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured Eγ. Deduced $T_{1/2}$ for two ^{35}Cl levels using Doppler Shift Attenuation lineshape analysis.

^{35}Cl Levels

E(level)†	J^{π}‡	$T_{1/2}$#	Comments
0	3/2$^+$		
1219	1/2$^+$	201 fs *28*	$T_{1/2}$: Lifetime=0.29 ps *4* from 1973Wa10.
1762	5/2$^+$	374 fs *49*	$T_{1/2}$: Lifetime=0.54 ps *7* from 1973Wa10.

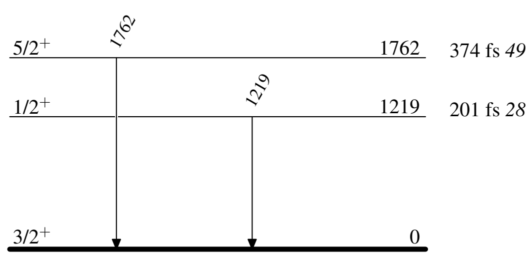
† From 1973Wa10.
‡ From the Adopted Levels.
\# From Doppler Shift Attenuation Method (DSAM) in 1973Wa10.

$\gamma(^{35}Cl)$

E_γ	E_i(level)	J^π_i	E_f	J^π_f
1219	1219	1/2$^+$	0	3/2$^+$
1762	1762	5/2$^+$	0	3/2$^+$

$^{2}H(^{34}S,n\gamma)$ 1973Wa10

Level Scheme



$^{35}_{17}Cl_{18}$